

Dylan Grosz

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SUMMARY

ML Scientist specializing in on-device speech and speaker recognition at billion-user scale. At Apple, own the end-to-end ML pipeline for Siri voice trigger and speaker verification – from post-training optimization and UX-aligned threshold tuning to automated evaluation and global locale deployment, driving 8–24x efficiency gains. Published researcher (AAAI, PAKDD; 90 citations) with a U.S. patent pending.

EXPERIENCE

Machine Learning Scientist

Sep 2021 – Present

Apple – Siri, Attention & Invocation

Seattle, WA

- Own the speaker recognition and voice trigger ML pipeline for “Hey Siri” / “Siri” and Recognize My Voice, serving **1B+ active devices** across **iPhone, Mac, HomePod, Apple TV, Apple Watch, and Vision Pro** in **40+ locales**
- Architected the team’s automated model evaluation and post-training tuning system, reducing release validation from **1 day (3 engineers) to under 1 hour (1 engineer)** and cutting asset deployment from **8 hours to 1 hour** – eliminating the primary bottleneck in the team’s release cadence
- Defined the methodology for UX-driven acceptance threshold optimization, applying Bayesian optimization (TPE) to jointly tune false-acceptance and false-rejection rates against product-level UX targets – replacing ad hoc manual tuning with a repeatable, auditable framework
- Drove the internationalization of speaker recognition from **7 to 40+ supported locales**, navigating per-language acoustic variability and hardware-specific constraints to enable Recognize My Voice in 20+ new markets
- Led the architecture migration from LSTM to Conformer-based speaker embeddings, systematically evaluating candidate architectures against Apple’s on-device compute, memory, and privacy constraints
- Led gender fairness analysis across demographic cohorts; pioneered LLM-powered automation for release workflows; established onboarding curriculum for new ML engineers

Software Engineer 1 / Machine Learning Engineer 2

Jul 2020 – Aug 2021

PayPal – SRE: Actionable Insights

San Jose, CA

- Invented a reinforcement learning approach to infrastructure autoscaling (**U.S. Patent Pending, Pub. No. US20220383324A1**) – the first ML-based autoscaling system deployed in PayPal’s SRE organization
- Designed and delivered PayPal’s SRE 200 ML education series, establishing the org’s first structured path for engineers to develop ML competency

Software Engineer Intern

Summers 2017 – 2019

PayPal

San Jose, CA

- Received rare **management impact award**; founded PayPal’s **Machine Learning Guild** (75+ members); co-founded the **Global Innovation Tournament**, a company-wide annual competition still running today
- Built **payment forecasting model** (random forests, RNNs, neural decomposition; $R^2 > 0.99$) and **Generalized Data Triage**, a notebook tool for automated dataset analysis

PUBLICATIONS, PATENT & PROJECTS

[GOOGLE SCHOLAR](#)

- U.S. Patent Pending (Pub. No. US20220383324A1) – “Dynamic Autoscaling of Server Resources Using Intelligent Demand Analytic Systems.” Filed 2022.
- **D. Grosz**, P. Conde-Céspedes. “Automatic Detection of Sexist Statements Commonly Used at the Workplace.” *PAKDD LDRC*, 2020. **53 citations**; dataset is a standard benchmark in hate speech detection.
- J. Lee, **D. Grosz**, B. Uzgent, et al. “Predicting Livelihood Indicators from Crowdsourced Imagery.” *AAAI*, 2021. **31 citations**.
- I. Fischer-Hwang, **D. Grosz**, X.E. Hu, et al. “Disarming Loaded Words: Addressing Gender Bias in Political Reporting.” *Computation+Journalism*, 2020.

EDUCATION

Stanford University

Class of 2020

B.S. Symbolic Systems (CS, Math, Statistics, Linguistics) – AI Concentration, Minor in Economics

GPA: 3.85

- ML Research Assistant, Stanford Sustainability & AI Lab and ISEP (Paris)
- Founded **The Stanford Daily @94305**, recruiting 55 students for a new data journalism section

TECHNICAL SKILLS

Core Expertise: On-Device ML, Speaker Recognition, Speech Processing, Voice Trigger, Model Evaluation, Deep Learning, NLP

Technical: Python, C++, SQL, PySpark, PyTorch, TensorFlow, scikit-learn, Kaldi, Bayesian Optimization, Conformer/Transformer

At Scale: Billion-device deployment, MLOps, multi-locale optimization, automated ML pipelines, CI/CD | Bilingual: English, French